

MARIAGIULIA MARTINELLI

Bioarchaeology | Palaeogenetics | Conservation Science

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PROFILE

My research follows the biological processes that act on human skeletal remains and heritage materials: taphonomy and bone diagenesis, microbial colonisation and biodeterioration of stone, and the preservation of ancient biomolecules. I work across **bioarchaeology**, **palaeogenetics** and **conservation science**, with a constant methodological interest in reproducible quantitative analysis: I write and openly release my own R pipelines, and carry results through to figures, reports and manuscripts. My most recent focus has been the multi-proxy reconstruction of human biocultural transitions through **ancient dental calculus** and **metagenomic classification**, combining molecular and microscopic approaches.

EDUCATION

Master's Degree in *Science and Technology for the Conservation of Cultural Heritage* (LM-11)

Sapienza University of Rome

09/2024 – graduating 21/10/2026

- Thesis title: *Conservation and Reassessment of an Overlooked Skeletal Collection: Historical and Bioarchaeological Study of Human Crania from Etruscan Territory Acquired by G. Sergi in the Early 20th Century.*
- Thesis supervisors: Dr. Ileana Micarelli, Prof. Mary Anne Tafuri, Prof. Laura Maria Michetti
- Programme delivered in Italian and English; admission conditional on assessed B2 English (CEFR).
- Student Representative for the Department of Environmental Biology (Bachelor, Master and PhD level) and for the LM-11 Master's Degree Class.

Erasmus+ Exchange Student

Stockholm University, Sweden

01/2026 – 06/2026

- Courses: *Palaeogenetics* (BL7069); *Modelling Tools for Environmental Scientific Studies* (MI4011); *Communicating Environmental Science* (MI4008).

Bachelor's Degree in *Tecnologie per la Conservazione e il Restauro dei Beni Culturali* (L-43)

Final grade: 110/110

Sapienza University of Rome

09/2021 – 12/2024

- Thesis title: *Studio dell'evoluzione del microbiota sulle pareti del Casino Algardi (Roma) in seguito ad interventi conservativi con prodotti sostenibili.*
- Thesis supervisors: Prof. Enea Gino Di Domenico, Dr. Chiara Alisi, Prof. Viviana Fonti, Prof. Claudio Chimenti
- Student Representative of the L-43 Degree Class.

RESEARCH EXPERIENCE

Research Internship at the *Centre for Palaeogenetics* (CPG)

Stockholm University / Sapienza University of Rome

04/2026 – 06/2026

- Analysed 250 ancient DNA libraries from Roman, Visigothic, Islamic and Medieval necropolises in Valencia (Spain), sampled from petrous bone, teeth and phalanges: elements with markedly different susceptibility to post-mortem alteration.

- Ran Kraken2 (GTDB) and KrakenUniq (MicrobialNT) classification pipelines as SLURM job arrays on the Dardel HPC cluster (PDC, KTH), resolving the endogenous signal from the soil-derived community that records post-mortem colonisation of bone.
 - Assessed cross-tool agreement (Procrustes, Spearman, PCoA on Bray–Curtis) and screened all samples for *Yersinia pestis*, *Mycobacterium tuberculosis* and *Tannerella forsythia*; delivered count tables, figures and a comparative report.
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Osteological Laboratory, *Tusculum* Project

EEHAR-CSIC (Escuela Española de Historia y Arqueología en Roma)

📅 12/2025 – 01/2026

- Catalogued and analysed medieval human remains from the Area G cathedral site (DG-ABAP no. 245): skeletal inventory and element identification, biological profile, and macroscopic recording of preservation, weathering and taphonomic alteration.
 - Managed the digitisation of the anthropological dataset, ensuring traceability between remains, records and photographic documentation.
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Research Internship at the *Palaeoanthropology and Bioarchaeology Laboratory*

Sapienza University of Rome

📅 07/2025 – 12/2025

- Bioanthropological study and critical re-evaluation of a human cranial collection acquired by Giuseppe Sergi (early 20th century): biological profile, dental palaeopathology, history of previous study, and systematic assessment of the state of conservation, distinguishing taphonomic damage from damage introduced during excavation and curation. The objective was to revitalise a neglected scientific heritage collection and propose new avenues for research.
 - Designed the reproducible R pipeline used for the collection (*Etruscan-Analysis*, released open source).
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Research Internship at the ENEA Research Centre, Casaccia (Rome)

ENEA Casaccia

📅 04/2024 – 12/2024

- Designed and executed independently an end-to-end metagenomics workflow on microbially mediated deterioration of a mineral substrate: field sampling and photographic documentation of biofilms on the historic walls of the *Casino del Bel Respiro*, genomic DNA extraction, PCR (16S rRNA, ITS), amplicon sequencing and phylogenetic classification.
 - Built the R pipeline from scratch (phyloseq, vegan, ggplot2): alpha- and beta-diversity, taxonomic composition and functional-group succession under sustainable conservation treatments. Presented as an oral contribution (UniPi, 2025) and a co-authored abstract.
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Research Internship: Taxonomic Identification and Iconographic Analysis

Sapienza University of Rome

📅 2022 – 2024

- Study of *The Four Elements* cycle by Jan Brueghel the Elder: bibliographic research, taxonomic classification of the species represented, and high-resolution photographic documentation within the 17th-century natural-history context.
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AWARDS AND RECOGNITIONS

Tutoring Fellowship (Cat. B1), Degree Programmes in *Scienze Applicate ai Beni Culturali*

Department of Environmental Biology, Sapienza University of Rome

📅 a.y. 2026/2027

- Awarded through public comparative selection (Call BT-B1 18/2026, 2 positions available), scored on degree marks, academic record and previous tutoring experience.
- 75 hours of teaching and laboratory support for the L-43 and LM-11 degree classes (supervision: Prof. M. Beccaccioli, Prof. L. Medeghini, Prof. A. Ciccola).

Merit Scholarship – Academic Excellence Track (Master's)

Sapienza University of Rome

📅 03/2026 – 08/2026

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.
- Qualifying project: ancient metagenomic analysis of Roman–Medieval archaeological samples, at the Centre for Palaeogenetics, Stockholm.

Merit Scholarship – Academic Excellence Track (Bachelor's)

Sapienza University of Rome

📅 04/2023 – 12/2024

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.
- Qualifying project: iconographic and taxonomic analysis of the *Four Elements* cycle by Jan Brueghel the Elder.

Student Assistantship, Earth Science Library

Sapienza University of Rome

📅 2023 – 2025

- Merit-based student collaboration grant awarded through competitive University ranking.

PUBLICATIONS



C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso

Abstract in the Proceedings of *The Omics & Heritage (O&H)*, Rome, Italy, 14–15 May 2024

Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights

MANUSCRIPTS IN PREPARATION



M. Martinelli, V. Fonti, P. Paganin, F. Tasso, C. Alisi

International Biodeterioration & Biodegradation (manuscript in preparation, 2026)

Microbial Biodiversity on the Casino del Bel Respiro Walls: Implications for Conservation Strategies



L. Morucci, A.A. Bastanza, M. Martinelli, S. Iorio

International Journal of Paleopathology (manuscript in preparation, 2026)

A Systematic Review of Analytical Methodologies in Bioarchaeology of Stress in Non-Adult Populations: Recommendations for the Next 50 Years



M. Martinelli, C. Chimenti

Manuscript in preparation, 2026

The Animal World in Jan Brueghel's Four Elements

ORAL CONTRIBUTIONS



Orizzonti Antichi tra Ordine Perduto e Nuove Armonie

Oral contribution *Glycyrrhiza glabra* vs Biofilm: nuove prospettive per la conservazione dei beni culturali

M. Martinelli, V. Fonti, F. Tasso, P. Paganin, C. Alisi; University of Pisa · 02–04/12/2025



The Omics & Heritage (O&H)

Co-author of oral contribution *Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights*

C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso; CNR, ISB · 14–15/05/2024

OPEN SOURCE PROJECTS

-  **Etruscan-Analysis: Bioarchaeological Pipeline (R)**
Pipeline for a historical cranial collection: record standardisation and auditing, anatomical heat maps of dental pathology and taphonomic damage, stress markers by age cohort, and interactive provenance mapping. *tidyverse*, *sf*, *leaflet*; ships with a mock dataset for reuse.
-  **Bioarchaeological-Systematic-Review (R)**
Framework for literature data mining and statistical analysis of osteological stress markers in non-adult populations: 28 stress indicators, co-occurrence matrices, chronological parsing of site intervals, and non-parametric hypothesis testing. Analytical backbone of the systematic review above.
-  **kraken2-krakenuniq-comparison (R, Shell)**
R and shell scripts for the Valencia aDNA comparison on the Dardel HPC cluster: SLURM job arrays, abundance-matrix construction, cross-tool validation and pathogen screening.
-  **CO₂ Time Machine (JavaScript)**
Interactive 3D globe for temporal and spatial mapping of environmental data. Developed for *Communicating Environmental Science* (MI4008), Stockholm University.
-  **scienze-abc (HTML)**
Archive and graphical prototype for the testimonial section of *Storie dietro Scienze ABC*, the science-outreach project of *Sapienza* (scienzeabcsapienza.it).
-  **Biofilm Metagenomics Workflow (Casino Algardi), to be released**
R pipeline for 16S rRNA and ITS amplicon analysis of biofilms on stone surfaces: phyloseq diversity analysis and functional-group classification.

SKILLS

-  **Osteology, Taphonomy & Conservation of Skeletal Remains**
Skeletal inventory and element identification · biological profile (age, sex, stature) · dental palaeopathology · macroscopic assessment of bone preservation, weathering and taphonomic alteration · selection of elements and sampling loci · state-of-conservation assessment and collection management · photographic documentation and record-specimen traceability
-  **Laboratory & Analytical Practice**
Molecular: genomic and ancient DNA extraction · PCR (16S rRNA, ITS) · amplicon sequencing · sterile technique and contamination control in low-copy-number contexts
Microscopy & diagnostics: optical and stereomicroscopy of heritage and biological materials · biofilm sampling and documentation
-  **Data Analysis, Statistics & Reporting**
R (*tidyverse*, *phyloseq*, *vegan*, *ggplot2*, *sf*, *leaflet*) · non-parametric testing · multivariate ordination (PCoA, Bray-Curtis, Procrustes) · publication-quality figures and tables · reproducible pipelines, Git/GitHub, $\LaTeX$ · written reports and manuscripts
-  **Bioinformatics & High-Performance Computing**
Kraken2 · KrakenUniq · read-count-based pathogen screening · SLURM job arrays on HPC (Dardel, PDC/KTH) · shell scripting · Linux command line
-  **Core Scientific Background**
General, organic and analytical chemistry (BSc L-43 / MSc LM-11) · human anatomy and skeletal biology (bioanthropology laboratory practice) · microbiology and biology of deterioration · biostatistics
-  **Communication, Organisation & Autonomy**
Independent management of research projects from sampling design to written report · elected student representative across three degree levels (L-43, LM-11, PhD) · oral contributions at national and international conferences · co-authorship and manuscript drafting · academic writing in Italian and English

LANGUAGES

-  **Italian**
Native speaker (C2). Full academic and professional use.



English

B2 (CEFR), assessed at admission to the LM-11 Master's programme, which is delivered in Italian and English. Full Erasmus+ semester at Stockholm University with courses, laboratory work and examinations in English; oral presentations at international conferences; scientific writing and manuscript preparation.

CONFERENCES, WORKSHOPS & WEBINARS



International Conference on Palaeogenomics (ICP2026)

Stockholm, Sweden · 23–26/06/2026



Webinar: *Bottleneck o Consanguineità: come si entra in Accademia*

GIGA (Gruppo Italiano Giovani Antropolog3) · 15/12/2025



Webinar: *Starter Pack per la ricerca: dalla scrittura della tesi ai primi congressi*

GIGA (Gruppo Italiano Giovani Antropolog3) · 26/11/2025



Seminar: *Nuove opportunità professionali per i biologi nel mondo dell'arte e dei beni culturali*

FNOB, FIB · 27/02/2025



Webinar: *Primi passi con R: introduzione al linguaggio per analisi statistiche*

Consortium GARR · 12/06/2024



SAGacia24: *Primo Simposio Arte che Guarda alla Scienza*

Sapienza University of Rome · 05/04/2024



Workshop: *Dagli Acetali, ai DES, ai FAME: i nuovi solventi green introdotti da YOCOCU nel settore dei Beni Culturali*

YOCOCU · 29/01/2024



Seminar: *La meteorologia moderna e il ruolo guida dell'Italia nel suo sviluppo storico*

CREA, Università della Tuscia · 29/11/2023

WORK EXPERIENCE

Library Assistant

Library of the Department of Earth Sciences, *Sapienza University of Rome*

📅 01/2023 – 01/2025

Private Science Tutor

Self-employed

📅 2021 – 2023

Tennis Instructor

Reschio Hotel Summer Camp

📅 06/2021 – 08/2021

CERTIFICATIONS



Open Badge: Student Representation

Issued by Bestr · 2025



Open Badge: Service Learning

Issued by Bestr · 2024



Course: *D.Lgs. 81/2008 Health and Safety Training*

KPMG Advisory · 27/02/2023



Certificate of Excellence: Localizing the Sustainable Development Goals (MUSTER)

Issued by Dr. Karl-Ludwig Schibel, President of the Fiera delle Utopie Concrete, Liceo Scientifico "Plinio il Giovane"
· 2018

REFERENCES

Dr. Ileana Micarelli

@ Sapienza University of Rome



Dr. Chiara Alisi

@ ENEA Casaccia



Prof. Teresa Rinaldi

@ Sapienza University of Rome



Prof. Mary Anne Tafuri

@ Sapienza University of Rome



Prof. Enea Gino Di Domenico

@ Istituti Fisioterapici Ospitalieri



Prof. Claudio Chimenti

@ Sapienza University of Rome



Autorizzo il trattamento dei dati personali contenuti nel mio curriculum vitae ai sensi del Regolamento UE 2016/679 (GDPR) e della normativa italiana vigente. Acconsento inoltre a che la presente candidatura sia presa in considerazione nei 12 mesi successivi alla selezione anche per ulteriori posizioni con profilo professionale simile.

Signature: Mariagiulia Martinelli

Date: August 26, 2026